

CBSE Class 10 Mathematics — Polynomials

— Solutions

1. **1.** Assertion (A): For polynomials $p(x) = x^2 - 1$ and $g(x) = x - 1$, the remainder is 0. Reason (R): According to the division algorithm, $p(x) = g(x)q(x) + r(x)$, where $r(x) = 0$ if $g(x)$ is a factor of $p(x)$.

Answer: (A)

Solution:

1. Check if $g(x)$ is a factor of $p(x)$ using division or factor theorem.
2. Since $p(1) = 1^2 - 1 = 0$, $x - 1$ is a factor.
3. The division algorithm states the relationship between dividend, divisor, quotient, and remainder.
4. Both statements are true and the reason explains the assertion.

Derivation:

$$\begin{aligned}p(x) &= (x - 1)(x + 1) + 0 \\ \implies r(x) &= 0\end{aligned}$$

Polynomials — division algorithm for polynomials

2. **2.** Assertion (A): On dividing $x^2 + 5x + 6$ by $x + 2$, the remainder is 0. Reason (R): The degree of the remainder is always equal to the degree of the divisor.

Answer: (C)

Solution:

1. Check remainder: $(-2)^2 + 5(-2) + 6 = 4 - 10 + 6 = 0$. Assertion is true.
2. The division algorithm states that the degree of the remainder $r(x)$ must be less than the degree of the divisor $g(x)$. Reason is false.

Derivation:

$$\deg(r(x)) < \deg(g(x))$$

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3. **3.** Assertion (A): When $p(x) = 2x^2 + 3x + 1$ is divided by $x + 1$, the quotient is $2x + 1$. Reason (R): The division algorithm $p(x) = g(x)q(x) + r(x)$ holds for all polynomials where $g(x) \neq 0$.

Answer: (A)

Solution:

1. Perform division: $(2x^2 + 3x + 1)/(x + 1) = (x + 1)(2x + 1) + 0$.
2. The quotient is indeed $2x + 1$. Assertion is true.
3. The division algorithm is a fundamental property of polynomials. Reason is true and explains the process.

Derivation:

$$2x^2 + 3x + 1 = (x + 1)(2x + 1) + 0$$

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4. **4.** Assertion (A): If $p(x) = x^3 - 3x^2 + 5x - 3$ is divided by $x - 1$, the remainder is 0.
Reason (R): If $p(a) = 0$, then $(x - a)$ is a factor of $p(x)$.

Answer: (A)

Solution:

1. Calculate $p(1) = 1^3 - 3(1)^2 + 5(1) - 3 = 1 - 3 + 5 - 3 = 0$. Assertion is true.
2. The Factor Theorem states that if $p(a) = 0$, then $x - a$ is a factor and the remainder is 0. Reason is true and explains the assertion.

Derivation:

$$p(1) = 1 - 3 + 5 - 3 = 0$$

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5. **5.** Assertion (A): If a polynomial $p(x)$ is divided by a linear polynomial $g(x)$, the remainder is a constant. Reason (R): The degree of a linear polynomial is 1.

Answer: (B)

Solution:

1. If $g(x)$ is linear, degree is 1. The remainder must have degree less than 1, which means degree 0 (a constant). Assertion is true.
2. The reason is also a true mathematical fact, but it describes the divisor, not the core relationship between the division algorithm and the remainder degree.

Derivation:

$$\begin{aligned} \deg(r(x)) &< \deg(g(x)) = 1 \\ \implies \deg(r(x)) &= 0 \end{aligned}$$

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6. **6.** On dividing the polynomial $p(x) = x^3 - 3x^2 + x + 2$ by a polynomial $g(x)$, the quotient and remainder were $(x - 2)$ and $(-2x + 4)$ respectively. Find $g(x)$.

Answer:

$$g(x) = x^2 - x + 1$$

Solution:

1. Use the division algorithm $p(x) = g(x) \cdot q(x) + r(x)$ to express $g(x) = \frac{p(x) - r(x)}{q(x)}$.
2. Substitute the given values, perform the subtraction $p(x) - r(x)$, and divide by $q(x)$.

Derivation:

$$g(x) = \frac{(x^3 - 3x^2 + x + 2) - (-2x + 4)}{x - 2} = \frac{x^3 - 3x^2 + 3x - 2}{x - 2} = x^2 - x + 1$$

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7. **7.** Check whether $g(x) = x^2 + 3x + 1$ is a factor of the polynomial $p(x) = 3x^4 + 5x^3 - 7x^2 + 2x + 2$ by applying the division algorithm.

Answer:

No, it is not a factor.

Solution:

1. Divide $p(x)$ by $g(x)$ using long division.
2. Observe the remainder; if the remainder is zero, it is a factor; otherwise, it is not.

Derivation:

Dividing $3x^4 + 5x^3 - 7x^2 + 2x + 2$ by $x^2 + 3x + 1$ yields a quotient of $3x^2 - 4x + 2$ and a remainder of 2. Since the

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8. **8.** If the polynomial $f(x) = x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $x^2 - 2x + k$, the remainder comes out to be $x + a$. Find the values of k and a .

Answer:

$$k = 5, a = -5$$

Solution:

1. Perform polynomial long division of $f(x)$ by $(x^2 - 2x + k)$.
2. Equate the resulting remainder expression to $(x + a)$ and solve for the constants.

Derivation:

*Dividing $x^4 - 6x^3 + 16x^2 - 25x + 10$ by $x^2 - 2x + k$ gives quotient $x^2 - 4x + (8 - k)$ and remainder $(2k - 10)x + 10 - 2k$.
 $\implies k = 5.5$ (re-evaluating: $2k - 10 = 1$ is incorrect, the remainder is $(2k - 10)x + 10 - 2k$)*

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9. **9.** Find the quotient and remainder when $p(x) = x^3 - x^2 + x - 1$ is divided by $g(x) = x - 1$.

Answer:

Quotient : $x^2 + 1$, Remainder : 0

Solution:

1. Perform long division of $x^3 - x^2 + x - 1$ by $x - 1$.
2. Write down the quotient and remainder obtained.

Derivation:

$x^3 - x^2 + x - 1 = x^2(x - 1) + 1(x - 1) = (x^2 + 1)(x - 1)$. Thus, quotient is $x^2 + 1$ and remainder is 0.

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10. 10. What must be subtracted from $p(x) = 8x^4 + 14x^3 - 2x^2 + 7x - 8$ so that the resulting polynomial is exactly divisible by $g(x) = 4x^2 + x - 2$?

Answer:

$$\text{Subtract } 14x - 10$$

Solution:

1. Divide $p(x)$ by $g(x)$ using long division.
2. Identify the remainder; the polynomial to be subtracted is equal to the remainder.

Derivation:

Performing long division : $(8x^4 + 14x^3 - 2x^2 + 7x - 8) \div (4x^2 + x - 2)$ results in a quotient $(2x^2 + 3x - 2)$ and

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11. 11. On dividing $x^3 - 3x^2 + x + 2$ by a polynomial $g(x)$, the quotient and remainder were $(x - 2)$ and $(-2x + 4)$ respectively. Find $g(x)$.

Answer:

$$g(x) = x^2 - x + 1$$

Solution:

1. Use the division algorithm: $\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$.
1. Substitute the given values: $x^3 - 3x^2 + x + 2 = g(x)(x - 2) + (-2x + 4)$.
2. Rearrange to find $g(x)(x - 2) = x^3 - 3x^2 + x + 2 + 2x - 4$.
3. Simplify and divide the resulting polynomial by $(x - 2)$.

Derivation:

$$\begin{aligned} g(x)(x - 2) &= x^3 - 3x^2 + 3x - 2 \\ \implies g(x) &= \frac{x^3 - 3x^2 + 3x - 2}{x - 2} = x^2 - x + 1 \end{aligned}$$

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12. 12. Find all the zeroes of the polynomial $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$, if you know that two of its zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Answer:

$$\text{Zeroes are } \sqrt{2}, -\sqrt{2}, 1, 0.5$$

Solution:

1. Since $\pm\sqrt{2}$ are zeroes, $(x - \sqrt{2})(x + \sqrt{2}) = x^2 - 2$ is a factor.
2. Divide $p(x)$ by $x^2 - 2$ to find the other quadratic factor.
3. Factorize the resulting quadratic quotient to find the remaining zeroes.

Derivation:

$$\frac{2x^4 - 3x^3 - 3x^2 + 6x - 2}{x^2 - 2} = 2x^2 - 3x + 1 = (2x - 1)(x - 1)$$

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- 13. 13.** What must be subtracted from $f(x) = x^4 + 2x^3 - 13x^2 - 12x + 21$ so that the resulting polynomial is exactly divisible by $g(x) = x^2 - 4x + 3$?

Answer:

$$-24x + 6$$

Solution:

1. Perform polynomial long division of $f(x)$ by $g(x)$.
2. Identify the remainder $r(x)$ obtained after division.
3. The value to be subtracted must be equal to the remainder $r(x)$.

Derivation:

$$f(x) = (x^2 - 4x + 3)(x^2 + 6x + 8) + (-24x + 6). \text{ Thus, subtract } (-24x + 6).$$

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- 14. 14.** If the polynomial $x^4 + x^3 + 8x^2 + ax + b$ is divisible by $x^2 + 1$, find the values of a and b .

Answer:

$$a = 1, b = 7$$

Solution:

1. Divide $x^4 + x^3 + 8x^2 + ax + b$ by $x^2 + 1$.
2. Perform the long division and determine the remainder in terms of a and b .
3. Since it is divisible, set the remainder equal to 0 and solve for coefficients.

Derivation:

$$\begin{aligned} \text{Remainder} &= x(a - 1) + (b - 7) = 0 \\ \implies a - 1 &= 0, b - 7 = 0 \end{aligned}$$

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- 15. 15.** Check whether $g(t) = t^2 - 3$ is a factor of $p(t) = 2t^4 + 3t^3 - 2t^2 - 9t - 12$ by applying the division algorithm.

Answer:

Yes, it is a factor

Solution:

1. Divide $p(t)$ by $g(t)$ using long division.
2. Observe the remainder.
3. If the remainder is 0, then $g(t)$ is a factor of $p(t)$.

Derivation:

$$\frac{2t^4 + 3t^3 - 2t^2 - 9t - 12}{t^2 - 3} = 2t^2 + 3t + 4, \text{ Remainder} = 0$$

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16. **16.** Obtain all the zeroes of the polynomial $p(x) = 2x^4 - 3x^3 - 3x^2 + 6x - 2$, if you know that two of its zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Answer:

$$1, \frac{1}{2}, \sqrt{2}, -\sqrt{2}$$

Solution:

1. Since $\sqrt{2}$ and $-\sqrt{2}$ are zeroes, $(x - \sqrt{2})(x + \sqrt{2}) = x^2 - 2$ is a factor of $p(x)$.
2. Divide $p(x)$ by $x^2 - 2$ using long division.
3. The quotient obtained is $2x^2 - 3x + 1$.
4. Factorize the quotient $2x^2 - 3x + 1$ by splitting the middle term.
5. Solve for the remaining zeroes by setting the factors equal to zero.

Derivation:

$$(2x^4 - 3x^3 - 3x^2 + 6x - 2) \div (x^2 - 2) = 2x^2 - 3x + 1. \text{ Factoring } 2x^2 - 2x - x + 1 = 2x(x - 1) - 1(x - 1) = (2x - 1)(x - 1)$$

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17. **17.** On dividing $x^3 - 3x^2 + x + 2$ by a polynomial $g(x)$, the quotient and remainder were $(x - 2)$ and $(-2x + 4)$ respectively. Find $g(x)$.

Answer:

$$g(x) = x^2 - x + 1$$

Solution:

1. Use the division algorithm formula: Dividend = (Divisor $\times$ Quotient) + Remainder.
2. Substitute the given values into $p(x) = g(x) \cdot q(x) + r(x)$.
3. Subtract the remainder $r(x)$ from the dividend $p(x)$.
4. Divide the resulting polynomial by the quotient $q(x)$ to find $g(x)$.
5. Perform the long division $g(x) = (x^3 - 3x^2 + 3x - 2)/(x - 2)$.

Derivation:

$$\begin{aligned} x^3 - 3x^2 + x + 2 &= g(x)(x - 2) + (-2x + 4) \\ \implies x^3 - 3x^2 + 3x - 2 &= g(x)(x - 2). \text{ Dividing } x^3 - 3x^2 + 3x - 2 \text{ by } x - 2 \text{ yields } x^2 - x + 1. \end{aligned}$$

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18. **18.** Find the values of a and b so that the polynomial $x^4 + x^3 + 8x^2 + ax + b$ is exactly divisible by $x^2 + 1$.

Answer:

$$a = 1, b = 7$$

Solution:

1. Perform polynomial long division of $x^4 + x^3 + 8x^2 + ax + b$ by $x^2 + 1$.
2. Determine the remainder in terms of a and b .
3. Since the polynomial is exactly divisible, the remainder must be zero.
4. Set the coefficients of the remainder polynomial to zero.
5. Solve the resulting linear equations to find a and b .

Derivation:

Dividing $x^4 + x^3 + 8x^2 + ax + b$ by $x^2 + 1$ gives quotient $x^2 + x + 7$ and remainder $(a - 1)x + (b - 7)$.

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19. **19.** Given that $x^2 + 1$ is a factor of $x^4 - 2x^3 + x^2 - 2x + 2$, use the division algorithm to find what must be subtracted from the polynomial so that the result is exactly divisible by $x^2 + 1$.

Answer:

$$\text{Subtracting } -2x + 2$$

Solution:

1. Divide the given polynomial by $x^2 + 1$.
2. Identify the remainder of the division.
3. According to the division algorithm, the remainder is the part that prevents the polynomial from being exactly divisible.
4. Conclude that subtracting the remainder makes the polynomial divisible.
5. Verify the result by subtracting the remainder from the dividend.

Derivation:

$(x^4 - 2x^3 + x^2 - 2x + 2) = (x^2 + 1)(x^2 - 2x) + (-2x + 2)$. To make it divisible, we must subtract the remainder.

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